

HBSC

Updated: 2026-09-03

Data Overview

It's a huge file:

```
## [1] 244097    120
```

With these columns:

HBSC	seqno_int	cluster	countryno	region
id1	id2	id3	id4	weight
adm	month	year	age	agecat
sex	grade	monthbirth	yearbirth	fasfamcar
fasbedroom	fascomputers	fasbathroom	fasdishwash	fasholidays
health	lifesat	headache	stomachache	backache
feellow	irritable	nervous	sleepdifficulty	dizzy
thinkbody	physact60	breakfastwd	breakfastwe	fruits_2
vegetables_2	sweets_2	softdrinks_2	fmeal	toothbr
timeexe	smokltm	smok30d_2	alc1tm	alc30d_2
drunkltm	drunk30d	cannabis1tm_2	cannabis30d_2	bodyweight
bodyheight	likeschool	schoolpressure	studtogether	studhelpful
studaccept	teacheraccept	teachercare	teachertrust	bulliedothers
beenbullied	cbulliedothers	cbeenbullied	fight12m	injured12m
friendhelp	friendcounton	friendshare	friendtalk	emconlfreq1
emconlfreq2	emconlfreq3	emconlfreq4	emconlpref1	emconlpref2
emconlpref3	emcsocmed1	emcsocmed2	emcsocmed3	emcsocmed4
emcsocmed5	emcsocmed6	emcsocmed7	emcsocmed8	emcsocmed9
hadsex	agesex	contraceptcondom	contraceptpill	countryborn
countrybornmo	countrybornfa	motherhome1	fatherhome1	stepmohome1
stepfahome1	fosterhome1	elsehome1_2	employfa	employmo
employnotfa	employnotmo	talkfather	talkstepfa	talkmother
talkstepmo	famhelp	famsup	famtalk	famdec
MBMI	IRFAS	IRRELFAS_LMH	IOTF4	oweight_who

So after selecting and grouping variables, create dat with :

```
## ## Individual Information
## * 'seqno_int'
## * 'countryno'
## * 'weight'
## * 'agecat'
## * 'sex'
## * 'MBMI'
```

```

## * 'IRFAS'
## * 'IRRELFAS_LMH'
## * 'oweight_who'
## * 'bodyweight'
## * 'bodyheight'
##
## ## Health Issues and Life Satisfaction
## * 'health'
## * 'lifesat'
## * 'headache'
## * 'stomachache'
## * 'backache'
## * 'feellow'
## * 'irritable'
## * 'nervous'
## * 'sleepdifficulty'
## * 'dizzy'
##
## ## Problematic Social Media Use (PMSU)
## * 'emcsocmed1'
## * 'emcsocmed2'
## * 'emcsocmed3'
## * 'emcsocmed4'
## * 'emcsocmed5'
## * 'emcsocmed6'
## * 'emcsocmed7'
## * 'emcsocmed8'
## * 'emcsocmed9'
##
## ## Bullying
## * 'beenbullied'
## * 'cbeenbullied'
##
## ## Communicates Online More Than Real Life
## * 'emconlfreq1'
## * 'emconlfreq2'
## * 'emconlfreq3'
## * 'emconlfreq4'
## * 'emconlpref1'
## * 'emconlpref2'
## * 'emconlpref3'
##
## ## Family Support
## * 'famhelp'
## * 'famsup'
## * 'famtalk'
## * 'famdec'
##
## ## Friends Support
## * 'friendhelp'
## * 'friendcounton'
## * 'friendshare'
## * 'friendtalk'
##

```

```

## ## School Support
## * 'likeschool'
## * 'schoolpressure'
## * 'studtogether'
## * 'studhelpful'
## * 'studaccept'
## * 'teacheraccept'
## * 'teachercare'
## * 'teachertrust'
##
## ## Communication with Parents
## * 'talkfather'
## * 'talkmother'
##
## ## Good/Bad Habits
## * 'physact60'
## * 'timeexe'
## * 'alc30d_2'

```

The number of missing values is relevant and need to address later:

##	variable	na_count	na_pct
## 1	oweight_who	54573	22%
## 2	MBMI	53311	22%
## 3	bodyweight	41128	17%
## 4	bodyheight	40036	16%
## 5	emcsocmed8	32878	13%
## 6	emcsocmed9	32855	13%
## 7	emcsocmed7	32597	13%
## 8	emcsocmed6	32544	13%
## 9	emcsocmed5	32443	13%
## 10	emcsocmed4	32428	13%
## 11	emcsocmed3	32009	13%
## 12	emcsocmed1	31973	13%
## 13	emcsocmed2	31659	13%
## 14	emconlpref3	31097	13%
## 15	emconlpref2	30586	13%
## 16	emconlpref1	29813	12%
## 17	emconlfreq2	29552	12%
## 18	emconlfreq4	29003	12%
## 19	emconlfreq3	28973	12%
## 20	emconlfreq1	26720	11%
## 21	famdec	20237	8%
## 22	famsup	20091	8%
## 23	famtalk	20019	8%
## 24	alc30d_2	19326	8%
## 25	famhelp	18896	8%
## 26	teachercare	18340	8%
## 27	teachertrust	17839	7%
## 28	talkfather	17491	7%
## 29	studaccept	17444	7%
## 30	beenbullied	16962	7%
## 31	teacheraccept	16764	7%
## 32	studhelpful	16746	7%

## 33	studtogether	16351	7%
## 34	talkmother	15521	6%
## 35	cbeenbullied	15084	6%
## 36	IRFAS	12599	5%
## 37	IRRELFAS_LMH	12599	5%
## 38	sleepdifficulty	10814	4%
## 39	friendshare	10813	4%
## 40	friendtalk	10808	4%
## 41	friendcounton	10374	4%
## 42	friendhelp	9744	4%
## 43	dizzy	8566	4%
## 44	feellow	6696	3%
## 45	backache	6543	3%
## 46	irritable	6539	3%
## 47	nervous	6525	3%
## 48	physact60	6215	3%
## 49	stomachache	5679	2%
## 50	schoolpressure	5486	2%
## 51	likeschool	5109	2%
## 52	timeexe	4799	2%
## 53	lifesat	4512	2%
## 54	headache	4505	2%
## 55	health	2577	1%
## 56	agecat	1516	1%
## 57	seqno_int	0	0%
## 58	countryno	0	0%
## 59	weight	0	0%
## 60	sex	0	0%

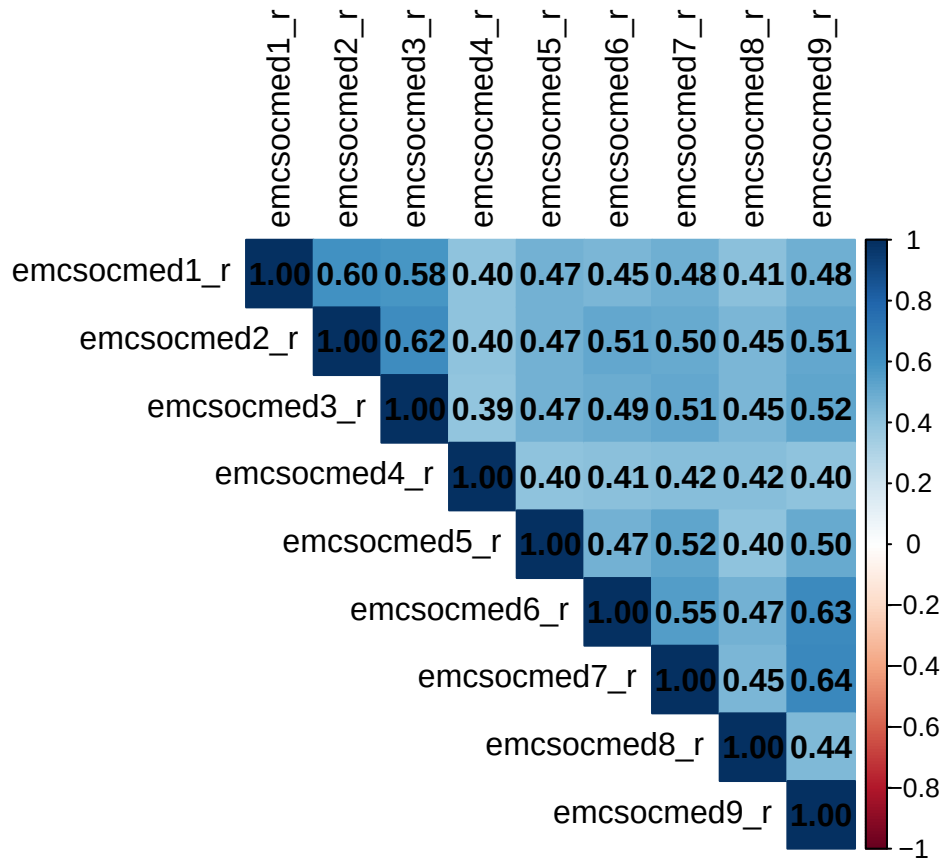
Now I explore all variables by group and feature engineer as appropriate.

[3] “Problematic Social Media Use (PMSU)”

9 binary variables are recoded to 0/1.

item	topic
emcsocmed1	Social media: I can't think of anything else
emcsocmed2	Social media: Dissatisfied cause want to spend more time
emcsocmed3	Social media: Felt bad cause want to play
emcsocmed4	Social media: Failed to spend less time
emcsocmed5	Social media: Neglected other activities
emcsocmed6	Social media: Had arguments because of use
emcsocmed7	Social media: Lied about amount of time spent
emcsocmed8	Social media: Use it to escape from negative feelings
emcsocmed9	Social media: Conflict with family because of use

And they are correlated.



These 9 vars are sumed and then binned to H/M/L where H(6-9), M(2-5) and L(0-1). Resulting in:

	1	2	3	NA
0	72474	0	0	0
1	36277	0	0	0
2	0	26066	0	0
3	0	19012	0	0
4	0	14908	0	0
5	0	10306	0	0
6	0	0	5905	0
7	0	0	3382	0
8	0	0	1878	0
9	0	0	2979	0
NA	0	0	0	50910

[6] “Family Support”

4 questions going from 1 Very strongly disagree to 7 Very strongly agree.

item	topic
famhelp	Family tries to help
famsup	Get emotional help from family
famtalk	Talk about problems with family

item	topic
famclose	Feel close to family

Summing them results in [4,28] and then grouped as H/M/L as L 4-11, M 12-19, H 20-28. Resulting in:

	1	2	3	NA
4	10939	0	0	0
5	1597	0	0	0
6	1452	0	0	0
7	1382	0	0	0
8	2477	0	0	0
9	1485	0	0	0
10	2274	0	0	0
11	1739	0	0	0
12	0	2655	0	0
13	0	2279	0	0
14	0	2745	0	0
15	0	2781	0	0
16	0	5747	0	0
17	0	3440	0	0
18	0	4455	0	0
19	0	4637	0	0
20	0	0	7739	0
21	0	0	5853	0
22	0	0	9240	0
23	0	0	7878	0
24	0	0	13474	0
25	0	0	11089	0
26	0	0	13090	0
27	0	0	15938	0
28	0	0	85359	0
NA	0	0	0	22353

- voy [1] “Individual Information”
[2] “Health Issues and Life Satisfaction”
[4] “Bullying”
[5] “Communicates Online More Than Real Life”
[7] “Friends Support”
[8] “School Support”
[9] “Communication with Parents”
[10] “Good/Bad Habits”